

Problem Set 14

Problem 12.5.2: (2 points)

(1) From the text (eqns. 12.5.23 and 12.5.24) we read off

$$J_x^{(1/2)} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_y^{(1/2)} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad J_z^{(1/2)} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

where the last, $J_z^{(1/2)}$, follows immediately from the definition of the $|j, m\rangle$ states.

Then

$$[J_x^{(1/2)}, J_y^{(1/2)}] = \frac{\hbar^2}{4} \left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\} = \frac{\hbar^2}{4} \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} = i\hbar J_z^{(1/2)}.$$

(2)

$$J_x^{(1)} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y^{(1)} = \frac{i\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_z^{(1)} = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

so

$$\begin{aligned} [J_x^{(1)}, J_y^{(1)}] &= \frac{i\hbar^2}{2} \left\{ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \right\} \\ &= \frac{i\hbar^2}{2} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} = i\hbar J_z^{(1)}. \end{aligned}$$

(3) From (12.5.21a,b), for $j = 3/2$ we have

$$\begin{aligned} \langle \tfrac{3}{2}, m' | J_x | \tfrac{3}{2}, m \rangle &= \frac{\hbar}{2} \left\{ \delta_{m', m+1} \sqrt{(\tfrac{3}{2} - m)(\tfrac{5}{2} + m)} + \delta_{m', m-1} \sqrt{(\tfrac{3}{2} + m)(\tfrac{5}{2} - m)} \right\}, \\ \langle \tfrac{3}{2}, m' | J_y | \tfrac{3}{2}, m \rangle &= \frac{\hbar}{2i} \left\{ \delta_{m', m+1} \sqrt{(\tfrac{3}{2} - m)(\tfrac{5}{2} + m)} - \delta_{m', m-1} \sqrt{(\tfrac{3}{2} + m)(\tfrac{5}{2} - m)} \right\}, \\ \langle \tfrac{3}{2}, m' | J_z | \tfrac{3}{2}, m \rangle &= \hbar m \delta_{m', m}. \end{aligned}$$

So, letting $m, m' \in \{\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\}$ gives

$$\begin{aligned} J_x^{(3/2)} &= \frac{\hbar}{2} \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}, & J_y^{(3/2)} &= \frac{i\hbar}{2} \begin{pmatrix} 0 & -\sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & -2 & 0 \\ 0 & 2 & 0 & -\sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}, \\ J_z^{(3/2)} &= \frac{\hbar}{2} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}. \end{aligned}$$

and so

$$\begin{aligned}
[J_x^{(3/2)}, J_y^{(3/2)}] &= \frac{i\hbar^2}{4} \left\{ \begin{pmatrix} 3 & 0 & -2\sqrt{3} & 0 \\ 0 & 1 & 0 & -2\sqrt{3} \\ 2\sqrt{3} & 0 & -1 & 0 \\ 0 & 2\sqrt{3} & 0 & -3 \end{pmatrix} - \begin{pmatrix} -3 & 0 & -2\sqrt{3} & 0 \\ 0 & -1 & 0 & -2\sqrt{3} \\ 2\sqrt{3} & 0 & 1 & 0 \\ 0 & 2\sqrt{3} & 0 & 3 \end{pmatrix} \right\} \\
&= \frac{i\hbar^2}{4} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -6 \end{pmatrix} = i\hbar J_z^{(3/2)}.
\end{aligned}$$

Problem 12.5.3: (2 points)

(1) By eqn. (12.5.21a,b),

$$\begin{aligned}
\langle jm|J_x|jm\rangle &= \frac{\hbar}{2} \{ \delta_{jj} \delta_{m,m+1}(\dots) + \delta_{jj} \delta_{m,m-1}(\dots) \} = 0, \\
\langle jm|J_y|jm\rangle &= \frac{\hbar}{2i} \{ \delta_{jj} \delta_{m,m+1}(\dots) - \delta_{jj} \delta_{m,m-1}(\dots) \} = 0,
\end{aligned}$$

since $\delta_{m,m\pm 1} = 0$.

(2) By rotational invariance about the z -axis, $\langle J_x^2 \rangle = \langle J_y^2 \rangle$. Since $J^2 - J_z^2 = J_x^2 + J_y^2$, therefore

$$\langle J_x^2 \rangle = \langle J_y^2 \rangle = \frac{1}{2} \langle J^2 - J_z^2 \rangle = \frac{1}{2} \langle jm|J^2 - J_z^2|jm\rangle = \frac{1}{2} (\hbar^2 j(j+1) - \hbar^2 m^2).$$

(3) $\Delta J_x = \sqrt{\langle J_x^2 \rangle - \langle J_x \rangle^2} = \sqrt{\frac{1}{2} \hbar^2 (j(j+1) - m^2)} = \Delta J_y$, and

$$\begin{aligned}
\langle J_x J_y \rangle &= \langle \frac{1}{2}(J_+ + J_-) \frac{1}{2i}(J_+ - J_-) \rangle = \frac{1}{4i} \langle J_+^2 - J_-^2 + J_- J_+ - J_+ J_- \rangle = \frac{1}{4i} \langle J_- J_+ - J_+ J_- \rangle \\
&= \frac{1}{4i} \langle jm| \left\{ J_- \left(\hbar \sqrt{(j-m)(j+m+1)} |j, m+1\rangle \right) - J_+ \left(\hbar \sqrt{(j+m)(j-m+1)} |j, m-1\rangle \right) \right\} \\
&= \frac{\hbar}{4i} \left\{ \sqrt{(j-m)(j+m+1)} \hbar \sqrt{(j+m+1)(j-m)} \langle jm|jm\rangle \right. \\
&\quad \left. - \sqrt{(j+m)(j-m+1)} \hbar \sqrt{(j-m+1)(j+m)} \langle jm|jm\rangle \right\} \\
&= \frac{\hbar^2}{4i} \{ (j-m)(j+m+1) - (j+m)(j-m+1) \} = i \frac{m\hbar^2}{2},
\end{aligned}$$

where we used at the end of the first line that $\langle J_{\pm}^2 \rangle = 0$ since they are either both raising or both lowering operators, so can only connect m with $m \pm 2$ states. Thus, we get

$$\Delta J_x \Delta J_y = \frac{1}{2} \hbar^2 [j(j+1) - m^2], \quad |\langle J_x J_y \rangle| = \frac{1}{2} |m| \hbar.$$

But $j \geq |m|$, implying $j(j+1) \geq |m|(|m|+1) = m^2 + |m|$, so $j(j+1) - m^2 \geq |m|$, which implies $\Delta J_x \Delta J_y \geq |\langle J_x J_y \rangle|$.

- (4) By the above derivation, the inequality is only saturated when $j = |m|$, which implies $m = \pm j$.

Problem 12.5.13: (2 points) $\psi = N(x + y + 2z)e^{-\alpha r}$.

- (1) From eqn. (12.5.35) we have

$$Y_1^m = -\frac{1}{2}\sqrt{\frac{6}{4\pi}}\sqrt{\frac{(m+1)!}{2(1-m)!}}e^{im\phi}(\sin\theta)^{-m}\frac{d^{1-m}(\sin^2\theta)}{d(\cos\theta)^{1-m}},$$

from which it follows that

$$\begin{aligned} Y_1^1 &= -\frac{1}{2}\sqrt{\frac{3}{2\pi}}e^{i\phi}(\sin\theta)^{-1}(\sin^2\theta) = -\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta e^{i\phi}, \\ Y_1^0 &= -\frac{1}{4}\sqrt{\frac{3}{\pi}}\frac{d(1-\cos^2\theta)}{d(\cos\theta)} = \left(\frac{3}{4\pi}\right)^{1/2}\cos\theta, \\ Y_1^{-1} &= -\frac{1}{4}\sqrt{\frac{3}{2\pi}}e^{-i\phi}(\sin\theta)\frac{d^2(1-\cos^2\theta)}{d(\cos\theta)^2} = \frac{1}{2}\sqrt{\frac{3}{2\pi}}e^{i\phi}\sin\theta\frac{d(\cos\theta)}{d(\cos\theta)} = +\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta e^{-i\phi}. \end{aligned}$$

In spherical coordinates $x \pm iy = r \sin\theta e^{\pm i\phi}$, and $z = r \cos\theta$, so

$$Y_1^{\pm 1} = \mp \left(\frac{3}{8\pi}\right)^{1/2} \frac{x \pm iy}{r}, \quad Y_1^0 = \left(\frac{3}{4\pi}\right)^{1/2} \frac{z}{r}.$$

- (2) Therefore

$$\begin{aligned} \psi &= N(x + y + 2z)e^{-\alpha r} \\ &= N\left\{\left(\frac{1-i}{2}\right)\left(\frac{x+iy}{r}\right) + \left(\frac{1+i}{2}\right)\left(\frac{x-iy}{r}\right) + 2\frac{z}{r}\right\}re^{-\alpha r} \\ &= N\left\{-\left(\frac{1-i}{2}\right)\left(\frac{8\pi}{3}\right)^{1/2}Y_1^1 + \left(\frac{1+i}{2}\right)\left(\frac{8\pi}{3}\right)^{1/2}Y_1^{-1} + 2\left(\frac{4\pi}{3}\right)^{1/2}Y_1^0\right\}re^{-\alpha r} \\ &= \mathcal{N}(r)\left\{-\frac{1-i}{\sqrt{2}}Y_1^1 + \frac{1+i}{\sqrt{2}}Y_1^{-1} + 2Y_1^0\right\}, \end{aligned}$$

where $\mathcal{N}(r) \equiv N(4\pi/3)^{1/2}re^{-\alpha r}$. Therefore

$$|\psi\rangle = |\mathcal{N}\rangle \left(-\frac{1-i}{\sqrt{2}}|1, 1\rangle + \frac{1+i}{\sqrt{2}}|1, -1\rangle + 2|1, 0\rangle\right).$$

So

$$\text{Prob}(\ell_z = 0) = \frac{|2|^2}{|2|^2 + \left|\frac{1+i}{\sqrt{2}}\right|^2 + \left|\frac{1-i}{\sqrt{2}}\right|^2} = \frac{4}{4 + 1 + 1} = \frac{2}{3},$$

$$\text{Prob}(\ell_z = \hbar) = \frac{\left|\frac{1-i}{\sqrt{2}}\right|^2}{|2|^2 + \left|\frac{1+i}{\sqrt{2}}\right|^2 + \left|\frac{1-i}{\sqrt{2}}\right|^2} = \frac{1}{4 + 1 + 1} = \frac{1}{6},$$

$$\text{Prob}(\ell_z = -\hbar) = \frac{\left|\frac{1+i}{\sqrt{2}}\right|^2}{|2|^2 + \left|\frac{1+i}{\sqrt{2}}\right|^2 + \left|\frac{1-i}{\sqrt{2}}\right|^2} = \frac{1}{4 + 1 + 1} = \frac{1}{6}.$$